

Ex. No: 1 Text Preprocessing Tasks

Implement Text Preprocessing Tasks such as Text Normalization, Text Cleaning, Tokenization, Stop word removal, Stemming and Lemmatization for the given Datasets

Datasets:

- 1. Brown Corpus
- 2. Reuters Corpus
- 3. Newsgroups Corpus
- 4. Universal Dependencies English Tree Bank

Aim

To implement various text preprocessing tasks including Text Normalization, Text Cleaning, Tokenization, Stop word removal, Stemming, and Lemmatization on standard NLP corpora in order to prepare raw text data for downstream natural language processing tasks.

Procedure/Pseudocode

- 1. Load the dataset (Brown, Reuters, Newsgroups, or UD English Tree Bank) using the appropriate library.
- 2. Text Normalization: Convert all text to lowercase and standardize Unicode characters.
- 3. Text Cleaning: Remove special characters, punctuation, numbers, and HTML tags using regular expressions.
- 4. Tokenization: Split cleaned text into individual tokens (words/sentences) using NLTK's `word_tokenize` or `sent_tokenize`.
- 5. Stop Word Removal: Load NLTK's English stop words list and filter out stop words from the token list.
- 6. Stemming: Apply Porter Stemmer or Lancaster Stemmer to reduce words to their root/stem form.
- 7. Lemmatization: Apply WordNet Lemmatizer to reduce words to their base dictionary form (lemma).
- 8. Display and compare the output at each preprocessing stage.

Libraries Used

- 1. nltk (Natural Language Toolkit) – for tokenization, stop word removal, stemming, lemmatization
- 2. re – for regular expression-based text cleaning
- 3. sklearn.datasets – for loading the 20 Newsgroups corpus
- 4. nltk.corpus – for Brown and Reuters corpora

- 5. conllu – for parsing Universal Dependencies English Tree Bank

Test Cases

- 1. Input: 'The quick brown foxes are jumping over the lazy dogs!' → After normalization: 'the quick brown foxes are jumping over the lazy dogs!' → After tokenization: ['the','quick','brown','foxes','are','jumping','over','the','lazy','dogs'] → After stop word removal: ['quick','brown','foxes','jumping','lazy','dogs'] → After stemming: ['quick','brown','fox','jump','lazi','dog'] → After lemmatization: ['quick','brown','fox','jump','lazy','dog']
- 2. Input: 'Running runs runner ran' → Stemming (Porter): ['run','run','runner','ran'] → Lemmatization: ['run','run','runner','run']
- 3. Input: Reuters sentence with HTML tags → Cleaned output strips tags and normalizes whitespace.

Results

The preprocessing pipeline successfully normalized, cleaned, tokenized, removed stop words, stemmed, and lemmatized text from all four corpora. Lemmatization produced linguistically more accurate base forms compared to stemming, while stemming was computationally faster. Token counts reduced significantly after stop word removal.